

Building the Budget

Your project's ten-step budget, in the worked example's tables. Fill each table from your Session 9 estimate and your Session 10 staffing plan.

Team: 9 **Project:** GPI: Gracias Por Invitar **Slice:** Event-goer features (7 stories, 37 points, 71 FP) **Date:** 10/04/2026
Status: Final

Work top to bottom. Steps 1 to 5 reuse the estimate and the staffing plan you already have; Steps 6 to 10 turn them into money. Every number you choose rather than measure is an assumption: label it and give it an owner. Price labor at loaded cost and productive hours, not salary and 160 hours. Every sourced number needs a source you can link.

BUILDING THE BUDGET

Estimate, Budget, Baseline: Three Different Things

	Estimate	Budget	Cost baseline
What it is	A forecast of effort and cost, with a range	The money leadership approves, including a cushion for surprises	The approved month-by-month spending plan you measure against
Set by	The project team	The sponsor (the executive who pays)	Built by the project manager, approved by the sponsor
Your project	394 to 852 hours = \$32,377 to \$70,014. Expected 603.67 hours = \$49,607 (traced to step 3 in this document)	Requested: \$58,868. Baseline: \$56,065. Cushion: \$2,803 (step 10)	\$56,065: the \$49,622 base build phased over 3 months (M1 \$17,820, M2 \$17,805, M3 \$13,997), plus \$6,443 contingency held back for named risks (step 10)

Your Session 9 estimate becomes a budget only when someone with authority approves it. The baseline is the version you are held to. Fill the third row last, from Step 10.

BUILDING THE BUDGET

Two Ways to Start a Budget

	Top-down	Bottom-up
How it starts	From a number someone wants, such as a fixed date or a fixed dollar amount.	From the list of work that has to be done, priced piece by piece.
Your example	Project Charter: 8 weeks × 50 team-hours = 400 hours × \$82.18 (per productive hour) ≈ \$32,870	Seven work packages (37 points, 71 function points): 603.67 hours, \$49,607 of labor plus \$15 non-labor = \$49,622
Use it for	A quick reality check. It tells you what leadership hopes for.	The budget you defend. It tells you what the work costs.

Build bottom-up, then set it beside the top-down number. The gap between the two is a finding: write it in Check 5.

YOUR BUDGET

Step 1: Write Down What the Cost Depends On

A cost-driving assumption is a statement that, if it turned out wrong, would change the cost. List 3 to 5. Write each one with an owner on your team.

Assumption	Why it drives cost	Owner
In the event-goer slice, we include 7 stories, which are 37 points and 71 function points. Any outside-source imports are out.	Everything outside the slice is unpriced. An import is 12 function points which equals about 108 hours.	Emmanuel Saenz, Project Manager (charter scope)
Organizers post events into GPI-created catalogue, and the team curates it first, like how the charter and Business Strategy have it.	Rules out imported work in the build and sets the curation hours in the operating cost	Oscar Vargas, Business Strategy Lead
Five people contributing 10 productive hours a week each, 50 team-hours a week, alongside coursework, for the whole build. Delivery work is split across developer, QA, UX and DevOps roles (65/17/10/8 percent, 545.17 delivery hours), and project management at 0.15 FTE is carved out of the 603.67 hours (58.5 hours).	Sets the 12.1-week expectation and the monthly spend. Fewer hours contributed extends the calendar, but total hours remain the same. Pricing project management on top would add about 58.5 hours (\$4,807).	Jazmin Huerta, market research lead
6 to 12 hours per function point, 70 hours per person-month for the comparable, with a 50 to 70 percent match and a 25 to 35 percent uplift. We have not measured any of these yet	Our whole range depends on these. After three sprints we'll replace them with our real numbers	Christian Lopez (Research Documentation Lead – Appendix owner)
Labor at \$51.36 an hour (ZipRecruiter) = \$106,829 a year. $\times 1.20$ burden = \$10,683 per FTE-month, or \$82.18 per productive hour. Tools and hosting stay on free tiers, since the charter has no money for paid licenses.	Every dollar amount is # of hours times this rate. A 1.4 burden would push labor up 17 percent, and paying to host the app would add a new cost.	Samuel Alderete (Portal Lead)

Notes:

- The 65/17/10/8 percentages are shares of the delivery hours (545.17), not of the full 603.67.
- The 0.15 is an FTE figure, not a percentage of hours. It means 15% of one full-time person each month, which is $0.15 \times 130 = 19.5$ hours a month, or 58.5 hours over 3 months.

This is how the hours are carved out:

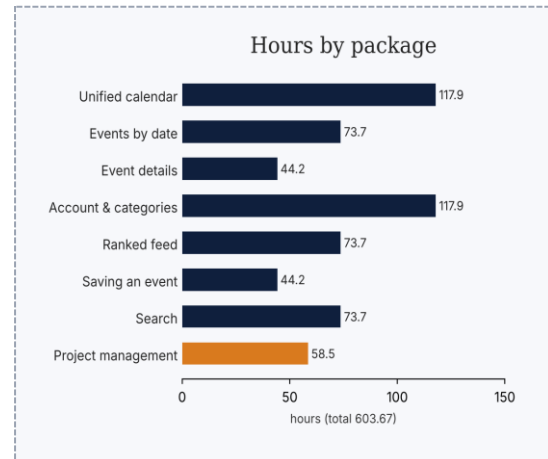
- Start with the total: 603.67 hours.
- Take the PM's share off first: $603.67 - 58.5 = 545.17$ hours left for delivery.
- Split those 545.17 hours 65/17/10/8 among the delivery roles.

The Session 9 estimate priced hours at \$61.63, a loaded rate per paid hour; this budget prices them per productive hour, \$82.18.

YOUR BUDGET

Step 2: Break the Work Into Packages

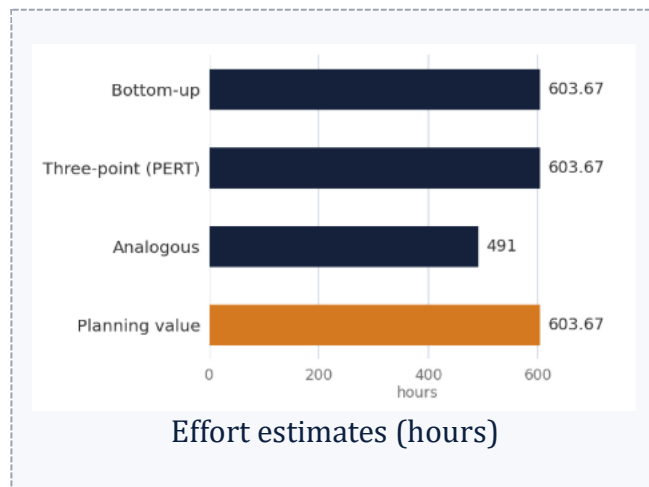
Work package	Role	Hours
Unified El Paso-Juarez calendar (8 pts)	Developers, UX, QA, DevOps (environment setup)	117.9
Upcoming events by date (5 pts)	Developers, UX, QA	73.7
Event time, place and details (3 pts)	Developers, UX, QA	44.2
Account and event categories (8 pts)	Developers, UX, QA	117.9
Feed ranked to categories (5 pts)	Developers, QA	73.7
Saving an event (3 pts)	Developers, QA	44.2
Search by name or keyword (5 pts)	Developers, QA, DevOps (deployment)	73.7
Project management (all packages)	Project manager (PM)	58.5
Total delivery effort	37 pts × 14.73 hrs + 58.5 PM	603.67



The rate becomes $545.17 \div 37 = 14.73$ hours per point, because 58.5 of the 603.67 hours go to project management first.

YOUR BUDGET

Step 3: Estimate Effort More Than One Way



Program Evaluation and Review Technique (PERT)

$$E = (O + 4M + P) \div 6$$

O = optimistic: 394 hours, if everything goes well

M = most likely: 594 hours

P = pessimistic: 852 hours, if real problems hit

$$E = 3,622 \div 6 = 603.67 \text{ hours}$$

Why E is above M: the bad case is farther from most likely than the good case is, so it pulls the answer up.

Spread = $(P - O) \div 6 = (852 - 394) \div 6 = 76.33$ hours: a typical miss. Step 9 reuses it.

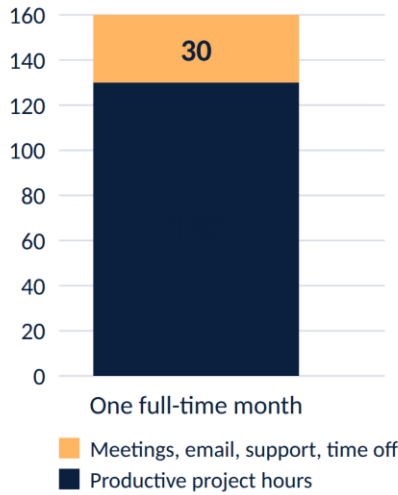
Method	Hours	What it starts from
Bottom-up (your packages)	603.67	The seven work packages in Step 2 at 14.73 hours per story point across 37 points, plus 58.5 project management hours. The package hours were distributed from the PERT expected value, so this line is not an independent check on PERT.

Three-point (PERT)	603.67	O 394 from the analogous low. P 852 from 71 function points at 12 hours each. M 594 from the midpoint of the two method ranges (623) and the midpoint of their two middles (565).
Analogous	491	Upcoming.org: one person over about 9 months at an assumed 70 hours per person-month, so 630 hours, scaled by a 50 to 70 percent feature match and a 25 to 35 percent coordination uplift. Range 394 to 595.
Planning value	603.67	The PERT expected value, which the bottom-up packages plus project management also sum to. This is the number carried into Step 4.

Do the methods land within 25 percent of each other? As a rule of thumb, if they differ by more than 25 percent, find out why before pricing anything. Write the reason here if they do: **Not needed, the spread is 23 percent. The planning value of 603.67 hours sits 22.9 percent above the 491 hour analogous midpoint, inside the 25 percent rule. Bottom-up and PERT are identical because the package hours were distributed from the PERT expected value, so analogous is the only independent comparison in the table.**

Step 4: Convert Hours to FTE-Months

160 nominal hours per month



Role	Hours	÷ 130	FTE-months
Developers	354.4	=	2.73
Quality Assurance (QA)	92.7	=	0.71
User Experience (UX)	54.5	=	0.42
DevOps	43.6	=	0.34
Delivery subtotal	545.17	=	4.19
Project manager (0.15 × 3 months)	58.5		0.45
Total	603.67		4.64

How to read it: one FTE-month is one person working full time for one month. About 80 percent of a month (130 of 160 hours) is productive project time; the rest goes to meetings, email, support and time off. Use your team's measured rate if you have one, and state the rate you assumed: 130 productive hours per FTE-month.

At 130 productive hours per FTE-month, 603.67 hours is 4.64 FTE-months of work. Our own Step 1 assumption is 10 productive hours per person per week, which is 50 team-hours a week, so those 4.64 FTE-months spread across about 12.1 weeks of calendar time. The role rows are rounded to two decimals; the total of 4.64 is taken from the unrounded values.

Any of our five members may fill the defined roles. Delivery hours are split by 65 percent developers, 17 percent QA, 10 percent UX and 8 percent DevOps, a team assumption owned by Jazmin Huerta (Step 1, row 3). Project management is priced at 0.15 FTE for 3 months, about 4.5 hours a week, and is carved out of the 603.67-hour planning value.

YOUR BUDGET

Step 5: Build the Staffing Plan by Month

Role	M1	M2	M3	M4	M5	M6	FTE-months
Project Manager	0.15	0.15	0.15	0	0	0	0.45
Developers	1.01	1.07	0.65	0	0	0	2.73
Quality Assurance	0.08	0.26	0.37	0	0	0	0.71
User experience	0.28	0.14	0	0	0	0	0.42
DevOps	0.15	0.05	0.14	0	0	0	0.34
Total FTE	1.67	1.67	1.31	0	0	0	4.64

Why the shape

For the UX, we have more hours dedicated to it in the beginning because the account, category, and calendar screens are designed first.

QA rises as features are finished and peaks in Month 3 for end-to-end testing.

DevOps peaks when the environment is set up in Month 1 and deployment in Month 3.

Developers carry out most of the work throughout, and the project manager keeps a steady 0.15 FTE.

Each month is capped at our capacity of 1.67 FTE (five people at 10 productive hours a week, 0.33 FTE each) and Month 3 is lighter because only about 0.8 of a month of work remains.

Check the total

All roles total 4.64 FTE-months the same as Step 4's FTE-months.

YOUR BUDGET

Step 6: Calculate the Loaded Labor Rate

\$106,829

blended base salary
per year
(\$51.36 x 2080 hours)

÷ 12

\$8902 per month

× 1.2

burden factor

\$10,683

per FTE-month

What is burden?

Costs an employer pays on top of salary: benefits, payroll taxes, equipment and workspace. Commonly used factors run about 1.25 to 1.4.

Rate	Value
Per nominal hour (÷ 160)	\$66.77
Per productive hour (÷ 130)	\$82.18

The 1.20 burden is a team assumption, below the 1.25 to 1.4 common range.

Pricing at the nominal-hour rate understates labor, because only 130 of 160 hours are productive.

YOUR BUDGET

Steps 6 and 7 Worked: Price the Labor and Everything Else

Labor: FTE-months × \$10,683

Role	FTE-months	Cost
Project manager	0.45	\$4,807
Developers	2.73	\$29,120
Quality assurance	0.71	\$7,616
User experience	0.42	\$4,480
DevOps	0.34	\$3,584
Labor total	4.64	\$49,607

Non-labor: everything else you buy

Item	How computed	Cost
Hosting: web service and database	Free tier during the build and the pilot (charter: no budget for paid licenses). A small tier would be about \$25 a month, an assumption to price from the vendor's published rates.	\$0
Domain name	One year, assumption	\$15
Transactional email for sign-up	Free tier at pilot volumes	\$0
Maps or geocoding for event places	Free tier at pilot volumes	\$0
Repository, CI, design tools	Free tiers	\$0
Organizer onboarding materials	Digital only: Labor, not a purchase	\$0
Non-labor total		\$15

Checklist: environments, tools, third parties, compliance, rollout.

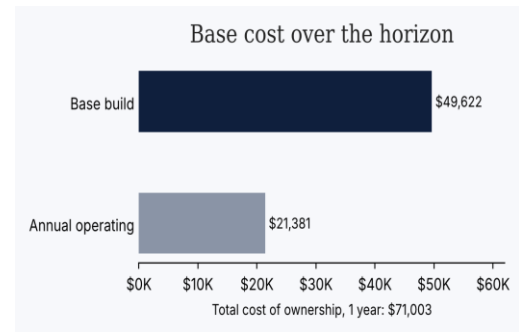
Base build estimate: \$49,607 labor + \$15 non-labor = \$49,622

YOUR BUDGET

Step 8: Price the Cost of Ownership

The build ends, but the system keeps costing money. It has to be hosted, fixed and supported every year.

Annual operating cost	How it was computed	Per year
Support and maintenance	5 team-hours a week × 52 × \$82.18	\$21,366
Production hosting	Free tier plus the domain	\$15
Licenses and services	Free tiers	\$0
Operating total	Sum of annual operation items	\$21,381

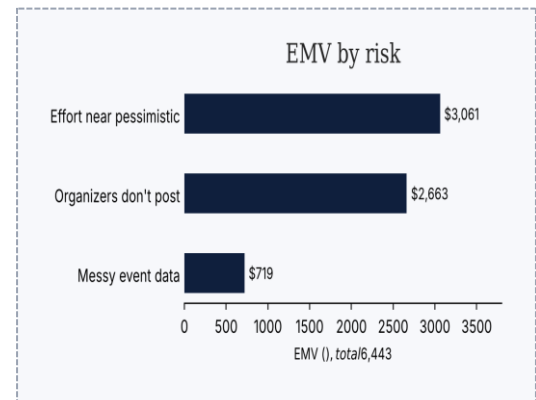


Why it belongs in the budget: over 1 year, operating cost (\$21,381) compared to the build (\$49,622). Total cost of ownership is \$71,003.

YOUR BUDGET

Step 9: Price the Risks (Contingency Reserve)

Risk	Chance	Impact	EMV
Effort lands near the pessimistic 852 hrs; our rates are not measured yet (+248.33 hrs)	15%	\$20,407	\$3,061
Organizers don't post, so we import events from outside sources (108 hrs)	30%	\$8,875	\$2,663
Uneven El Paso-Juarez event data needs a cleanup and review pass (35 hrs)	25%	\$2,876	\$719
Contingency reserve			\$6,443



How to read it: expected monetary value (EMV) = chance × impact. It is an average, not a prediction: a risk either costs about its impact or nothing, and the reserve covers the expected total. Take your top three risks at least.

Cross-check: One PERT spread from Step 3 × your productive-hour rate = **\$6,273** (76.33 hrs × \$82.18) of effort risk alone. Is the reserve at least that large? Yes, \$6,443 is \$170 more.

YOUR BUDGET

Step 9: Two Reserves

	Contingency reserve	Management reserve
Covers	Risks you have named (known unknowns)	Surprises nobody has named (unknown unknowns)
Inside the cost baseline?	Yes	No
Controlled by	Project manager	Sponsor or leadership
Sized by	The risk table (sum of EMV)	Organizational policy, here 5 percent

\$6,443

contingency, inside the baseline

\$2,803

management reserve: 5 percent of the \$56,065 baseline

Baseline = base build + contingency
\$56,065 = 49,622 + 6,443

A named risk happens

Organizers don't post and the team has to import events (risk 2, about \$8875). The project manager spends contingency without asking.

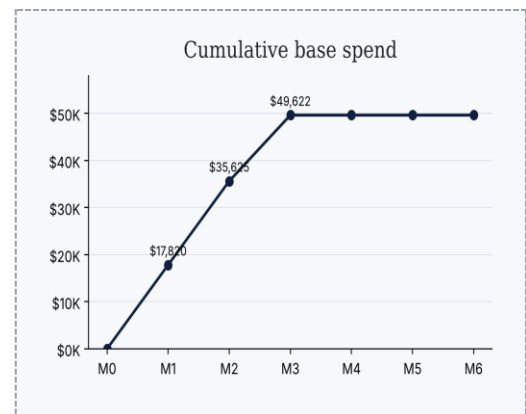
A surprise nobody listed

A teammate has to drop the course mid-build, and handing off their unfinished work costs hours nobody planned for. The project manager asks the sponsor to release management reserve.

YOUR BUDGET

Step 10: Time-Phase the Budget

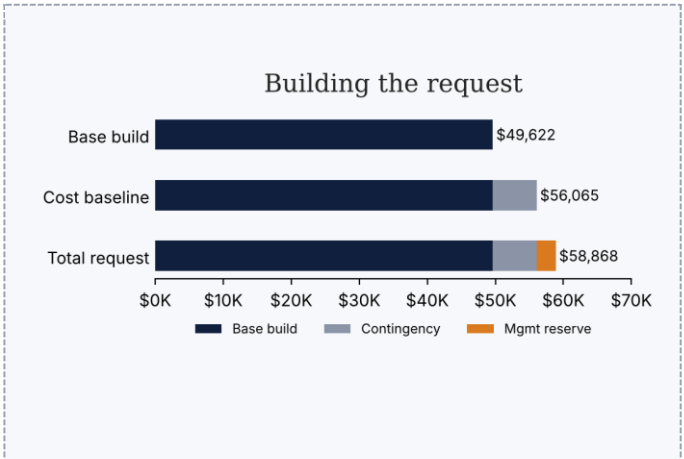
Month	Labor	Non-labor	Total	Cumulative
1	\$17,805	\$15	\$17,820	\$17,820
2	\$17,805	\$0	\$17,805	\$35,625
3	\$13,997	\$0	\$13,997	\$49,622
4	\$0	\$0	\$0	\$49,622
5	\$0	\$0	\$0	\$49,622
6	\$0	\$0	\$0	\$49,622



Time-phasing places each cost in the month it is spent. Labor = FTE that month × **\$10,683** per FTE-month (Step 6), using Step 5's unrounded FTE of 1.667, 1.667 and 1.310. The \$15 domain is bought in month 1. Contingency is held back until a named risk occurs, so this shows the base build only.

YOUR BUDGET

Step 10: The Full Budget Request



Line	Amount
Labor	\$49,607
Non-labor	\$15
Base build estimate	\$49,622
Contingency reserve (EMV)	\$6,443
Cost baseline	\$56,065
Management reserve (5%)	\$2,803
Total budget request	\$58,868
Operating cost (separate line)	\$21,381 / year

How to read it: each line stacks on the last. Base build is what you expect. Contingency covers named risks. Management reserve covers the unnamed.

YOUR BUDGET

Step 10: Five Sanity Checks

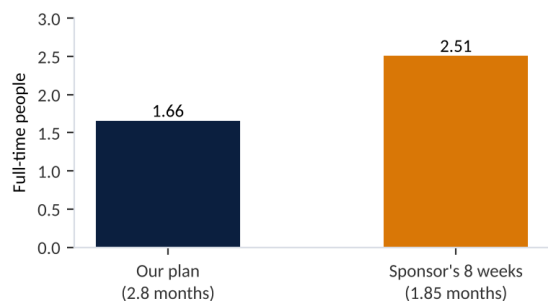
- ✓ **Totals match:** Add the staffing plan by month and compare it to Step 4. Ours: 4.64 FTE-months by month (1.667 + 1.667 + 1.310) = 4.64 FTE-months in Step 4.
- ✓ **Labor share is plausible:** Divide labor by the base build. Ours: $\$49,607 \div \$49,622 = 99.97$ percent. Nearly all cost is people because every tool and service stays on a free tier.
- ✓ **Estimates agree:** Compare the effort estimates. Ours: 491 to 603.67 hours, within 23 percent.
- ✓ **Reserves trace to risks:** Every contingency dollar links to a named risk in the Step 9 table. Ours: $\$3,061 + \$2,663 + \$719 = \$6,443$, all three named in the Step 9 table.
- ✓ **Top-down comparison:** Put the sponsor's number next to yours and explain the gap. Sponsor's number: 400 hours (8 weeks $\times$ 50 team-hours), about \$32,870 of labor. Yours: 603.67 hours, \$49,607 of labor. The gap is 203.67 hours, or \$16,737, about 51 percent more. Eight weeks at our 50 team-hours a week only fits 400 hours, but the seven stories in the slice price at 603.67 hours, which takes about 12.1 weeks at that pace. To close the gap either the slice gets smaller or the timeline gets longer.

YOUR BUDGET

Check 5 in Action: The Sponsor's Number

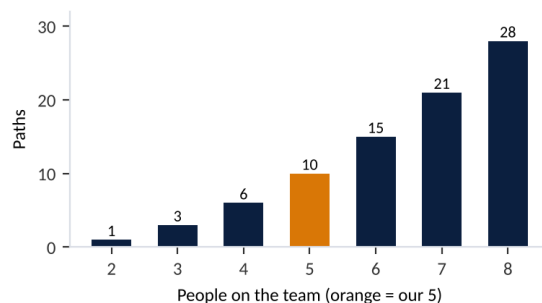
What did the sponsor ask for, in time or money? 8 weeks at 50 team-hours a week, so 400 hours (about \$32,870). Our budget says the work is 4.64 FTE-months.

People needed for the FTE-months



$4.64 \text{ FTE-months} \div 2.8 \text{ months} = 1.66 \text{ people}$. $4.64 \text{ FTE-months} \div 1.85 \text{ months (the sponsor's time)} = 2.51 \text{ people}$.

Communication paths, $n(n - 1) \div 2$



Every pair of people is a path where a misunderstanding can happen. 5 people means 10 of them.